F72-SR-001
415V REAGENTS SWITCHROOM

F72-SW-072
415V REAGENTS 415V MOTOR CONTROL CENTRE

11 kV FROM HV SWITCHBOARD
F23-SW-016
2000-F23-SLD-EL-10002

F72-TR-072
TRANSFORMER ROOM

F72-TR-072
11/0.433kV
1600kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
2500/ 5A
2.5VA
5P20
2500/ 1A
2.5VA
5P20
2500/ 5A
2.5VA
C10.5
00F-01
00F-08
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
AFD
ACB
2500A
2A
24VDC CONTROL POWER SUPPLY
NOTE 7
125A
160A
SD
NOTE 6
E-SHUT
F72SR001-LCS
NOTE 12
CONTINUED ON SHT 2
2000-F72-SLD-EL-10202

415V 3NE~ 50Hz 2500A 65kA for 1sec

2F-01 20A 2A 32A 70A EOL
0.75kW
F72-AG-658
SLAKED LIME DISCHARGE HOPPER 1 AGITATOR
2000-F72-SCM-EL-10668

2F-05 20A 2A 32A 70A EOL
0.75kW
F72-AG-659
SLAKED LIME DISCHARGE HOPPER 2 AGITATOR
2000-F72-SCM-EL-10669

1F-05 20A 2A 32A 70A EOL JB
3.7kW
F72-AT-411
LIME SILO 1 ACTIVATOR
2000-F72-SCM-EL-10651

1F-09 20A 2A 32A 70A EOL JB
3.7kW
F72-AT-412
LIME SILO 2 ACTIVATOR
2000-F72-SCM-EL-10652

5F-06 250A 2A 32A 100/1A
75kW
F72-BW-412
LIME DELIVERY BLOWER
2000-F72-SCM-EL-10665

5F-01 xxxA xxxA 2A 32A 100/1A
FUTURE VSD
75kW
F72-BW-413
LIME DELIVERY BLOWER (FUTURE)
2000-F72-SCM-EL-xxxxx

2F-09 20A 2A 32A 70A EOL
5.5kW
F72-FS-411
LIME SILO 1 SCREW FEEDER
2000-F72-SCM-EL-10654

2F-13 20A 2A 32A 70A EOL
5.5kW
F72-FS-412
LIME SILO 2 SCREW FEEDER
2000-F72-SCM-EL-10656

630A FEEDER
TO F72-SW-172
2000-F72-SLD-EL-10211

1R-13 630A 2A 32A 250A
5F-15 20A 2A 32A 70A EOL JB
2.2kW
F72-FV-411A
LIME SILO 1 VIBRATING CONVEYOR DRIVE 1
2000-F72-SCM-EL-10655
2.2kW
F72-FV-411B
LIME SILO 1 VIBRATING CONVEYOR DRIVE 2
2000-F72-SCM-EL-10655

5F-20 20A 2A 32A 70A EOL JB
2.2kW
F72-FV-412A
LIME SILO 2 VIBRATING CONVEYOR DRIVE 1
2000-F72-SCM-EL-10663
2.2kW
F72-FV-412B
LIME SILO 2 VIBRATING CONVEYOR DRIVE 2
2000-F72-SCM-EL-10663

3F-01 20A 2A 32A 70A
2.5kW
F72-CR-660
LIME SLAKING MILL BALL KIBBLE HOIST
2000-F72-SCM-EL-10664

Approved
Approved as Noted
Not Approved
For Information Only

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F72-SW-072 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F72-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- JB — JUNCTION BOX
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F72-SLD-EL-10201_Rev2

F72-SW-072 415V MCC - single line diagram sheet 1